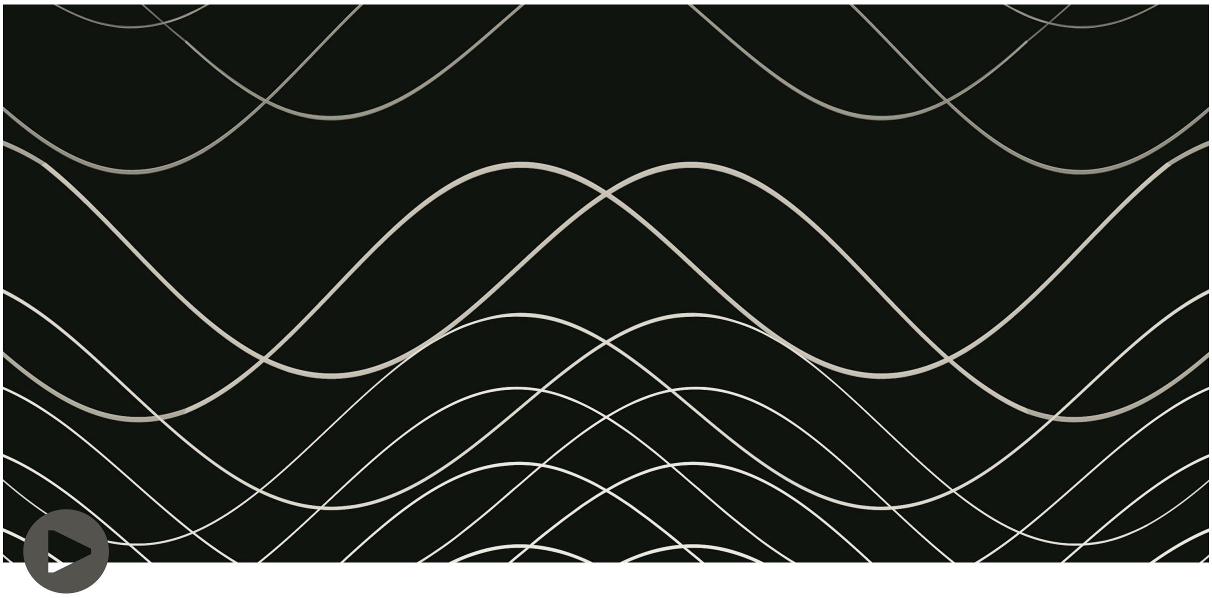

Research & Design for Hyperreal Audio: A Study on Human Perception in Immersive Soundscapes

https://drive.google.com/file/d/1PTpqpzcgR6kwyUgYsSvLFj1ay5IJMQd5/view?usp=drive_link

Instructions to listen to the sound experiment.

Please close your eyes and keep your head still for the whole time while liste ning to this aaudio recording.

Please use good quality headphones or earphones (preferrably closed-back headphones such as Beyerdynamic DT-770 Pro, Seinheiser HD 25 or Audio-Technica ATH-M50x) while listening to the audio.

https://drive.google.com/drive/u/0/folders/1lyrpl66ySgR5tAVDuQc9MwWDFbbp-E24

Project Overview

This Master's thesis is a rigorous, qualitative investigation into the psychoacoustics of immersion, specifically how human perception of reality is altered in VR/AR-ready auditory environments. The project demonstrates advanced proficiency in spatial audio production techniques (Ambisonics and Binaural) combined with crucial, user-centric design insights into the "hyperreal" state—the point where a user cannot distinguish between real and simulated soundscapes.

Key Demonstrations: Technical Proficiency

This project proves hands-on capability across the modern immersive audio pipeline:

- **Spatial Audio Production:** Designed and produced two distinct, high-fidelity auditory scenes using **Ambisonics** (Scene 2) and **Binaural** (Scene 1) recording and rendering techniques.
- **Professional Toolchain:** Expertise in signal processing and mixing within a Digital Audio Workstation (DAW) using dedicated spatial audio tools:
    - **DAW:** Reaper.
    - **Spatialization:** IEM Binaural Decoder, Dear VR Pro (used to accurately place virtual sound sources along the X-Y-Z axis).
- **Hardware and Format Management:** Experienced in handling multi-channel Ambisonics (B-Format/AmbiX) and Binaural files, utilizing specialized equipment (Zoom H3-VR, Roland CS-10EM) and working with Head-Related Transfer Functions (HRTFs) for realistic 3D sound rendering.
- **Scientific Rigor:** Employed a controlled, within-subjects research design across two physical locations (in-situ and ex-situ) to isolate variables and validate design principles.

Core Research Insights: The Psychology of Immersive Design

The qualitative findings, derived from thematic analysis of 24 semi-structured interviews, provide a scientific basis for design decisions in adaptive and immersive audio environments.

1. The Immediacy-Hypermediacy Dynamic (Immersion Control)

The study confirmed the principles of Bolter and Grusin's New Media theories, showing exactly what drives or breaks immersion:

- **Complete Immediacy (Full Immersion):** Achieved when visual cues were removed (Eyes Closed scenario) or when sound events were placed in the **far-field**. In these conditions, participants were unable to distinguish the audio scene from physical reality, successfully entering a "hyperreal" state.
  - ***Design Takeaway:*** Removing conflicting sensory information and utilizing distant sound layering is the most reliable path to achieving a profound sense of presence.
- **Hypermediacy (Awareness of the Medium):** Occurred when visual cues conflicted with the audio (Eyes Open scenario) or when sounds were placed in the **near-field** (within 1 meter). In these cases, participants became consciously aware they were listening to a recording, which broke the illusion.
  - ***Design Takeaway:*** Near-field sounds and front-facing events require the highest fidelity and contextual integration; otherwise, they are likely to break the user's perception of realism.

2. The Role of Memory and Context (Designing for the Hyperreal)

Findings using Gregory's Top-Down theory of perception provide a blueprint for creating psychologically convincing soundscapes:

- **Memory-Driven Realism:** Participants achieved the hyperreal state *only* when the spatial sounds in the virtual environment matched their **past experiences and memories** of the physical location. A strong personal schema for the location was the primary trigger for true immersion.
- **Context-Aware Triggers:** Identical sounds were perceived differently based on the **context of the physical playback environment**. For example, a knocking sound in the audio was perceived as a real-world threat in a room that actually had a window, but as an unrealistic sound in an open-air park.
- **Adaptive Audio Necessity:** The hyperreal state decreased with repeated exposure to the same sound events, as the brain "stabilized" its perception.

    - ***Design Takeaway:*** Immersive sound design, especially in long-form experiences, must incorporate adaptive, procedural elements (AI-driven variation) to prevent the brain from recognizing patterns and losing the illusion of reality.

Value Proposition for AI / VR / AR Sound Roles

This research demonstrates a rare combination of technical skill and deep UX insight. I don't just know *how* to mix spatial audio; I understand the **cognitive mechanisms** that determine if the mix is successful.

My work directly validates the need for:

- **Contextual Audio Systems:** Designing sound logic that accounts for a user's physical environment and prior experience (a critical component of AR audio).
- **Advanced Near-Field Rendering:** Prioritizing resources for near-field audio events to ensure high-fidelity, convincing sound localization that does not trigger a hypermediacy break.
- **Qualitative Validation:** Using thematic analysis as a valuable feedback loop for training adaptive sound engines to create and maintain the illusion of reality.

Hear the Experiment

A comprehensive write-up of the methodology, findings, and future work is available in the full thesis document.

[Link to Thesis PDF / Full Write-up]

[Link to Binaural/Ambisonics Audio Samples]

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Listen with headphones for a full 3D experience.